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Semester: 4
Subject Name: DBMS

WORKSHEET 4

AIM: To understand and apply SQL conditional logic using the MOD (%) operator to classify numerical data. The experiment focuses on analyzing employee salary values and determining whether they are odd or even, thereby improving data analysis and decision-making capabilities using SQL queries.

S/W Requirement:
- Database Management System: PostgreSQL / Oracle Database Express Edition
- Database Administration Tool: pgAdmin

OBJECTIVES:

- To understand the concept of conditional logic in SQL queries.
- To apply the MOD (%) operator to determine odd and even values.
- To analyze employee salary data using SQL conditions.
- To improve query writing skills for data filtering and classification.

PROBLEM STATEMENT:

Develop and execute SQL queries that demonstrate conditional logic using the MOD (%) operator. The program should analyze employee salary values and categorize them as **odd** or **even** using SQL queries.

1. PROBLEM STATEMENT – DISPLAY ALL EMPLOYEE DETAILS

Write an SQL query to create an employee table and display all employee records.

PROGRAM:

```
CREATE TABLE employee (
   emp_id NUMBER,
   emp_name VARCHAR2(50),
   salary NUMBER
```

);

INSERT INTO employee VALUES (101, 'Rahul', 25000);

INSERT INTO employee VALUES (102, 'Amit', 32001);

INSERT INTO employee VALUES (103, 'Neha', 45000);

INSERT INTO employee VALUES (104, 'Priya', 27555);

INSERT INTO employee VALUES (105, 'Rohan', 30000);

SELECT * FROM employee;

2. PROBLEM STATEMENT – DISPLAY EMPLOYEES WITH EVEN SALARY

Write an SQL query to display employees whose salary values are even numbers using the MOD operator.

PROGRAM:

DECLARE

SELECT emp_id, emp_name, salary

FROM employee

WHERE MOD(salary, 2) = 0;

3. PROBLEM STATEMENT – DISPLAY EMPLOYEES WITH ODD SALARY

Write an SQL query to display employees whose salary values are odd numbers using the MOD operator.

PROGRAM:

SELECT emp_id, emp_name, salary

FROM employee

WHERE MOD(salary, 2) = 1;

4. PROBLEM STATEMENT – CLASSIFY SALARY AS ODD OR EVEN

Write an SQL query to classify each employee salary as Odd or Even using a CASE statement.

PROGRAM:

```
SELECT emp_id,
    emp_name,
    salary,
    CASE
      WHEN MOD(salary,2)=0 THEN 'Even Salary'
      ELSE 'Odd Salary'
    END AS salary_type
FROM employee;
```

LEARNING OUTCOMES:

1. Understood how conditional logic can be applied in SQL queries.
2. Learned the use of the MOD (%) operator to determine odd and even values.
3. Gained knowledge about filtering records using conditional expressions.
4. Implemented CASE statements for classification of data.
5. Improved SQL query writing and logical analysis skills.

OUTPUT :

Navigator Files

My Schema

Tables

Search objects

EMPLOYEE

[SQL Worksheet]*

```sql
DROP TABLE employee;

CREATE TABLE employee (
    emp_id NUMBER,
    emp_name VARCHAR2(50),
    salary NUMBER
);

INSERT INTO employee VALUES (101, 'Rahul', 25000);
INSERT INTO employee VALUES (102, 'Amit', 32001);
INSERT INTO employee VALUES (103, 'Neha', 45000);
INSERT INTO employee VALUES (104, 'Priya', 27555);
INSERT INTO employee VALUES (105, 'Rohan', 30000);

SELECT * FROM employee;
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.006 seconds

	EMP_ID	EMP_NAME	SALARY
1	101	Rahul	25000
2	102	Amit	32001
3	103	Neha	45000
4	104	Priya	27555
5	105	Rohan	30000

[SQL Worksheet]*

```sql
    salary NUMBER
);

INSERT INTO employee VALUES (101, 'Rahul', 25000);
INSERT INTO employee VALUES (102, 'Amit', 32001);
INSERT INTO employee VALUES (103, 'Neha', 45000);
INSERT INTO employee VALUES (104, 'Priya', 27555);
INSERT INTO employee VALUES (105, 'Rohan', 30000);

SELECT * FROM employee;

SELECT emp_id, emp_name, salary
FROM employee
WHERE MOD(salary,2)=0;
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.007 seconds

	EMP_ID	EMP_NAME	SALARY
1	101	Rahul	25000
2	103	Neha	45000
3	105	Rohan	30000

[SQL Worksheet]*

```sql
INSERT INTO employee VALUES (102, 'Amit', 32001);
INSERT INTO employee VALUES (103, 'Neha', 45000);
INSERT INTO employee VALUES (104, 'Priya', 27555);
INSERT INTO employee VALUES (105, 'Rohan', 30000);

SELECT * FROM employee;

SELECT emp_id, emp_name, salary
FROM employee
WHERE MOD(salary,2)=0;

SELECT emp_id, emp_name, salary
FROM employee
WHERE MOD(salary,2)=1;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.004 seconds

	EMP_ID	EMP_NAME	SALARY
1	102	Amit	32001
2	104	Priya	27555

[SQL Worksheet]*

```sql
SELECT emp_id, emp_name, salary
FROM employee
WHERE MOD(salary,2)=0;

SELECT emp_id, emp_name, salary
FROM employee
WHERE MOD(salary,2)=1;

SELECT emp_id, emp_name, salary,
CASE
WHEN MOD(salary,2)=0 THEN 'Even Salary'
ELSE 'Odd Salary'
END AS salary_type
FROM employee;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.086 seconds

	EMP_ID	EMP_NAME	SALARY	SALARY_TYPE
1	101	Rahul	25000	Even Salary
2	102	Amit	32001	Odd Salary
3	103	Neha	45000	Even Salary
4	104	Priya	27555	Odd Salary
5	105	Rohan	30000	Even Salary

CONCLUSION:

This experiment helped in understanding the use of conditional logic in SQL queries using the MOD operator. By analyzing employee salary data and classifying it as odd or even, the experiment demonstrated how SQL can be used effectively for data analysis and logical decision-making.